知识点：第2章

信道的最大传输速率 ---计算题

- 波特（baud)、比特率（bit per second, bps)
- Signal-to-noise ratio S/N
- 分贝（DB) = $10\log_{10}S/N$
- **Nyquist 定理(无噪声信道):**

  Max速率（b/s) = $2H\log_2 V$

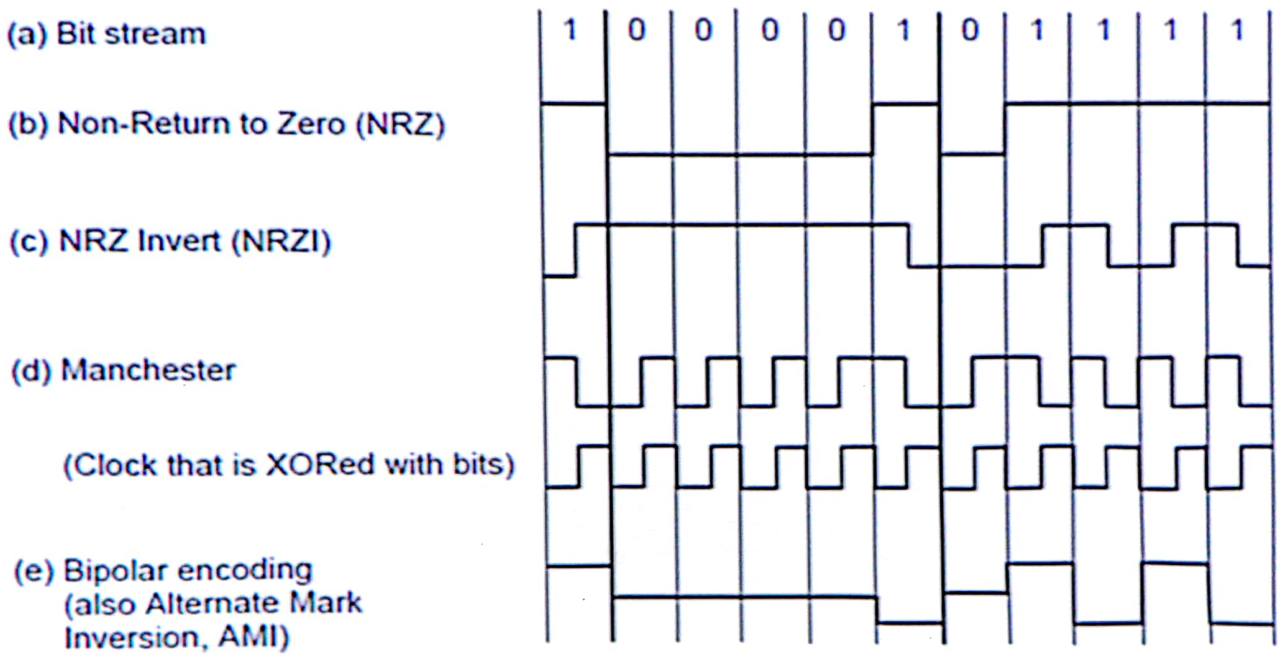
  --**H**:bandwidth of a low-pass filter

  --**V**:discrete levels of a signal
- **Shannon 定理(有噪声信道):**

  Max速率(b/s)= $H\log_2(1+S/N)$

知识点：第2章

- Baseband Transmission: Manchester
  - NRZ, NRZI, Manchester, Bipolar or AMI （see Next Page）
- Passband Transmission:
  - Amplitude shift keying
  - Frequency shift keying
  - Phase shift keying
- FDM\TDM\CDM
- PSTN：Local loop、Trunk、Siwtching的定义
- Local Loop: Modems 计算, ADSL and Wireless
- Trunks and Multiplexing： FDM,TDM,WDM
- Switching: Circuit/Message/Packet switching
- CDMA 计算


(a) Bit stream
1 0 0 0 0 1 0 1 1 1 1
(b) Non-Return to Zero (NRZ)
(c) NRZ Invert (NRZI)
(d) Manchester
(Clock that is XORed with bits)
(e) Bipolar encoding
(also Alternate Mark
Inversion, AMI)

知识点：第2章

- CDMA 计算， example：

A CDMA receiver gets the following chips: (+1 +1 -1 +3 +1 +1 -3 +1).
Assuming the chip sequences defined in following:

Station A: (-1 +1 -1 +1 +1 +1 -1 -1)
Station B: (-1 -1 -1 +1 +1 -1 +1 +1)
Station C: (-1 +1 -1 -1 -1 -1 +1 -1)
Station D: (-1 -1 +1 -1 +1 +1 +1 -1)

Which stations transmitted, and which bits did each one send?

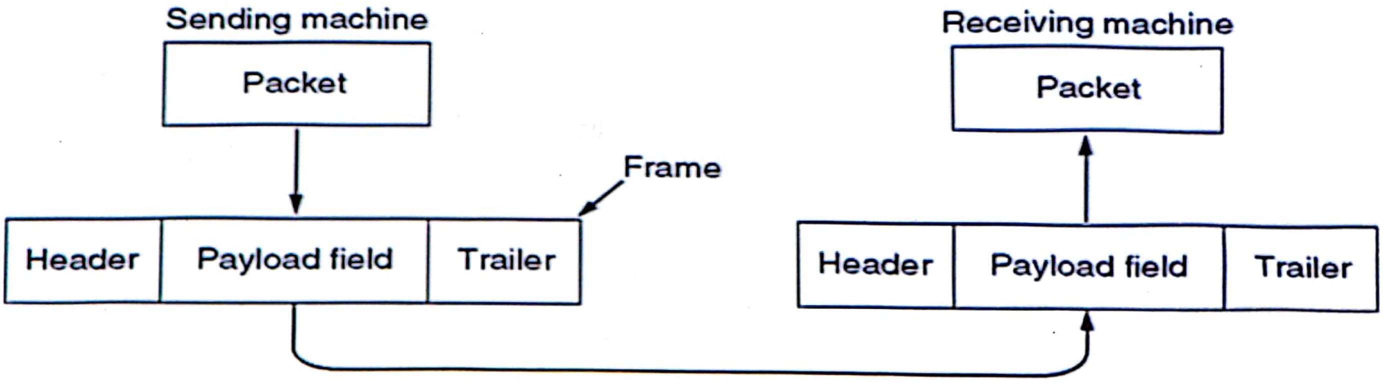
Answer:

SA*S = (-1+1+1+3+1+1+3-1)/8=1 SA transmitted a bit 1

Question: SA*S=1 ,SB*S=?, SC*S=?, SD*S=?

知识点：第3章

- Services Provided to the Network Layer
- Framing
- Error Control
- Flow Control
- Relationship between packets and frames:



知识点：第3章

- Framing：
  - Byte count、Flag bytes with byte stuffing、Flag bits with bit stuffing、Physical layer coding violations

(a) 0 1 1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 1 0

(b) 0 1 1 0 1 1 1 1 1 0 1 1 1 1 1 0 1 1 1 1 1 0 1 0 0 1 0

Stuffed bits

(c) 0 1 1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 1 0

- **Flow Control**
  - Feedback-based flow control
  - Rate-based flow control

知识点：第3章

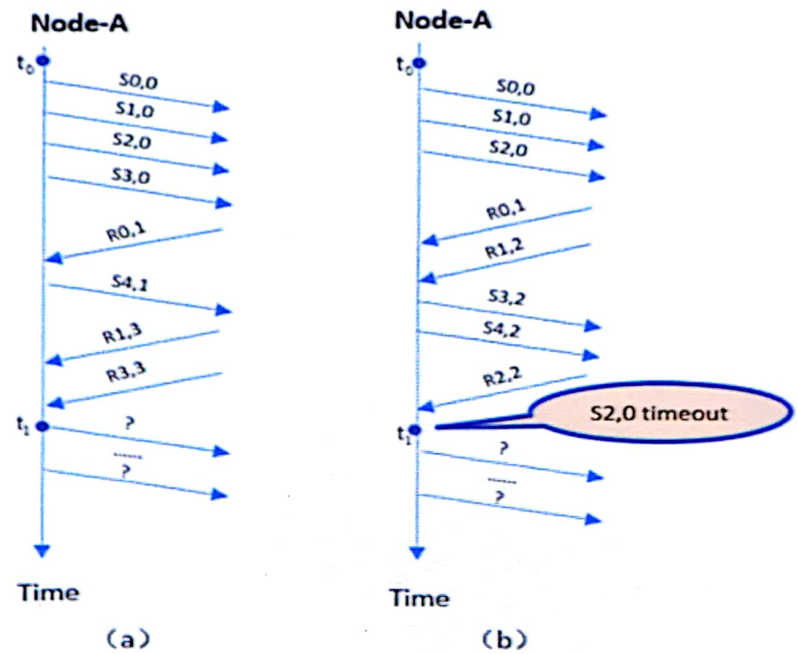
- Error-Correcting Codes（纠错码）
  - Hamming codes，Binary convolutional codes，Reed-Solomon codes，Low-Density Parity Check codes.
  - codeword（码字）
  - Hamming distance (10001001,10110001=>00111000,3)
  - parity bit
  - 纠正d比特错
    - 需距离为2d+1比特的编码
- Error-Detecting Codes（检错码）
  - polynomial code（cyclic redundancy code，CRC code）
  - 检测d比特错
    - 需距离为d+1比特的编码
  - 多项式编码
    - （循环冗余码，CRC码）
    - 尾数的计算

知识点：第3章

- **A Simplex Stop-and-Wait Protocol（一种单工的停-等协议）**
  - Error-Free Channel
- **A Simplex Protocol for a Noisy Channel（有噪声信道的单工协）**
  - Noisy Channel
  - PAR:Positive Acknowledgement with Retransmission
  - ARQ:Automatic Repeat reQuest
- **Sliding Window Protocols**
  - Piggybacking、 sending window、 receiving window
  - Line utilization (线路利用率) = l/(l+bR)，If the channel capacity is b bits/sec, the frame size l bits, and the round-trip propagation time R sec
  - Go Back N
    - 最多允许有MAX_SEQ帧未被确认
  - Selective Repeat
    - 窗口大小为W<=(MAX_SEQ+1)/2
- **Examples: HDLS、PPP**

Line utilization (线路利用率)

- This technique is known as pipelining. If the channel capacity is *b* bits/sec, the frame size *l* bits, and the round-trip propagation time *R* sec, the time required to transmit a single frame is l/b sec. After the last bit of a data frame has been sent, there is a delay of R/2 before that bit arrives at the receiver and another delay of at least R/2 for the acknowledgement to come back, for a total delay of R. In stop-and-wait the line is busy for l/b and idle for R, giving
- **Line utilization = *l/(l+bR)* *100%** （**= l/b /(l/b+R)** ）
  （单帧，不考虑ACK发送时间）
- **Line utilization = nl/b /(2*l/b+R) *100% [Go Back-N]**


Node-A
t_0
S0,0
S1,0
S2,0
S3,0
R0,1
S4,1
R1,3
R3,3
t_1
?
......
?
Time
Node-A
t_0
S0,0
S1,0
S2,0
R0,1
R1,2
S3,2
S4,2
R2,2
S2,0 timeout
t_1
?
......
?
Time

(a)　　　　(b)

知识点：第4章

- The categories of networks based on Transmission Technology
  - **Point-to-point connections（点到点连接）**
  - **Broadcast channels（广播信道）**
- Medium Access Control
- CSMA 仔细研究下面各种协议的特点：
  - ALOHA：Pure、Slotted basic idea
  - CSMA： **1-persistent、Nonpersistent、p-persistent**
  - CSMA/CD： IEEE 802.3(Ethernet)
- WLAN： CSMA/CA， RTS/CTS
  - Hidden station problem、 Exposed station problem 含义和特点
    - The problem of a station not being able to detect a potential competitor for the medium because the competitor is too far away is called the **hidden terminal problem**.
  - 802.11

知识点：第4章

- **Ethernet：**
  - Cables、DIX/802.3、classic/switched、Thin/thick/twisted pair/fiber
  - Manchester Encoding
  - CSMA/CD with Binary Exponential Backoff
  - delay=k*2τ, 2τ=51.2 μs
  - after i collisions, a random number k between 0 and 2^i-1 is chosen.
  - After 10 collisions have been reached, the randomization interval is frozen at a maximum of 1023 slots. After 16 collisions, the controller throws in the towel and reports failure back to the computer
  - maximum frame size:1518 bytes
  - minimum frame size: 64 bytes

知识点：第4章

- 硬件地址又称为物理地址，或MAC 地址, 48 bit
- Hub and Switch
- Collision Domain： In a hub, all stations are in the same collision domain; while in a switch, each port is a collision domain
- Fast Ethernet\ Gigabit Ethernet\ 10 Gigabit Ethernet
- Retrospective on Ethernet
- 802.11 : CSMA/CA DCF/PCF、RTS/CTS/ACK/NAV

知识点：第4章

- Bridge
  - **Use of Bridges**
  - Bridges from 802.x to 802.y
  - **Learning Bridges: Transparent,** hash table, flooding algorithm, **backward learning**, dynamic topologie, routing procedure for an incoming frame
  - **Parallel bridges,** Spanning Tree Bridges
  - **Repeaters, Hubs, Bridges, Switches, Routers and Gateways**
  - Virtual LANs based on Port\MAC\IP, 802.1Q
- **Collison domain**
- **Broadcast domain**

知识点：第4章

（**5 points***)* Consider building a CSMA/CD network running at 1 Gbps over a 1-km cable with no repeaters. The signal speed in the cable is 200,000 km/sec. What is the minimum frame size?

(Note: Please write the answer in answer sheet)

Answer: For a 1-km cable, the one-way propagation time is 5 msec, so 2t = 10 msec. To make CSMA/CD work, it must be impossible to transmit an entire frame in this interval. At 1 Gbps, all frames shorter than 10,000 bits can be completely transmitted in under 10 msec, so the minimum frame size is 10,000 bits or 1250 bytes.

评分标准：（1）计算出 one-way 传播时间 5msec 得 1分

（2）计算出 round trip 传播时间 2t=10msec 得 2分

（3）列出计算公式： 10ms/1000*1Gbps =10000bits 得4分

10,000 bits or 1250 bytes 得 5 分

知识点：第5章

- Network Layer Design Issues
- Routing Algorithms
- Congestion Control Algorithms
- Quality of Service
- Internetworking
- The Network Layer in the Internet ）

知识点：第5章

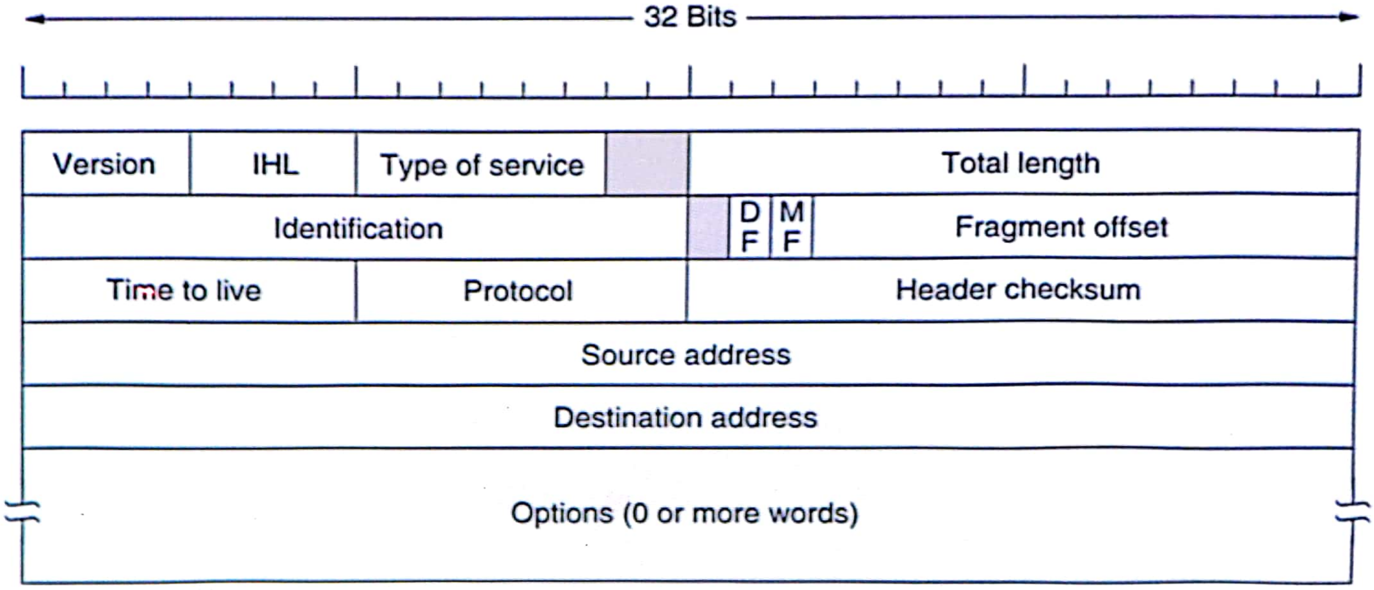
- ROUTING ALGORITHMS
  - ➢ Distance Vector Routing: **count-to-infinity problem**
    - ➢ **RIP\BGP**
  - ➢ Link State Routing: IS-IS, OSPF
- A router performs two tasks:
  - – **Forwarding**
  - – **Routing**

知识点：第5章

- **Congestion：causes (拥塞原因)**
- **Congestion control and flow control**
- QoS： A **flow** is a stream of packets from a source to a destination， four primary parameters：
  - **Reliability, Delay (延迟), Jitter (抖动), Bandwidth (带宽)**
- **MPLS、 Tunneling、Fragmentation**
- **AS， IGP， EGP，**

知识点：第5章

- The IPv4 (Internet Protocol) header



知识点：第5章

- IP address: network number and host number
  - Prefixes
  - Subnets (division),
  - CIDR(mergement),
  - Classful and Special Addressing
  - NAT
- IP address assignments
- supernet（超网）vs subnet
- Longest matching prefix
- Classful: A、B、C

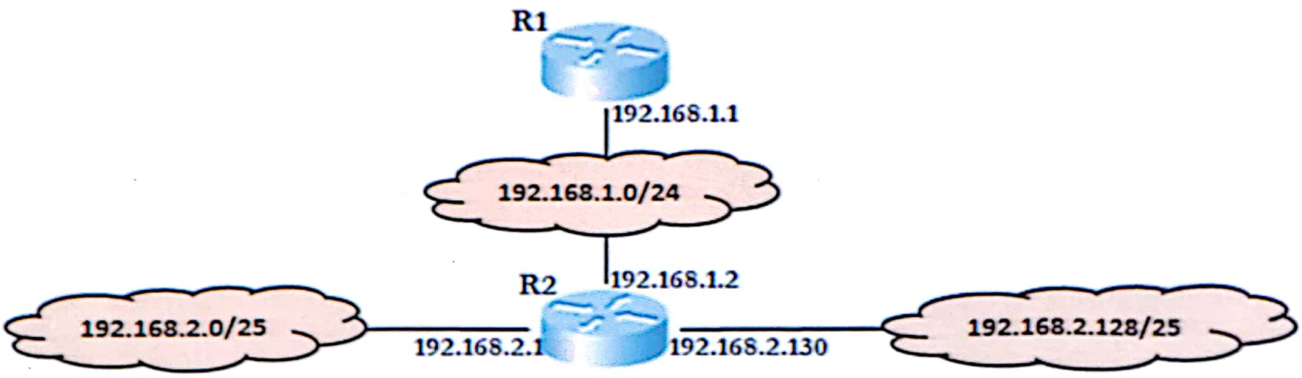
知识点：第5章

- **private address**
  - **10.0.0.0 到 10.255.255.255**
  - **172.16.0.0 到 172.31.255.255**
  - **192.168.0.0 到 192.168.255.255**
- the valid host range for a subnet: 192.168.10.16, mask 255.255.255.240?

  answer: 192.168.10.17 through 192.168.10.30
- NAT
- **IPv6, 128bit addresses, 地址表示**
- **ICMP、ARP、RARP、DHCP、BOOTP 特点**

知识点：第5章

- Routing table， route aggregation



知识点：第5章

- AS、IGP（RIP、OSPF）、EGP（BGP）
- Encapsulation protocol for OSPF protocol：IP
- Encapsulation protocol for RIP protocol：UDP

知识点：第5章

- 实验：
  - Cmd：Traceroute\ping\netcoonfig\netstate
  - How to config: static/RIP/OSPF/BGP
  - Cables，straight-through/cross-over
  - 配置交换机、路由器的各种命令
  - Routing table (how to understand)
  - 各种接口，local loop

知识点：第6章

- The Transport Service (传输服务)
- Elements of Transport Protocols (传输协议的若干问题)
- A Simple Transport Protocol (简单传输协议)
- The Internet Transport Protocols (Internet传输协议): UDP
- The Internet Transport Protocols (Internet传输协议): TCP
- Performance Issues (性能问题)

知识点：第6章

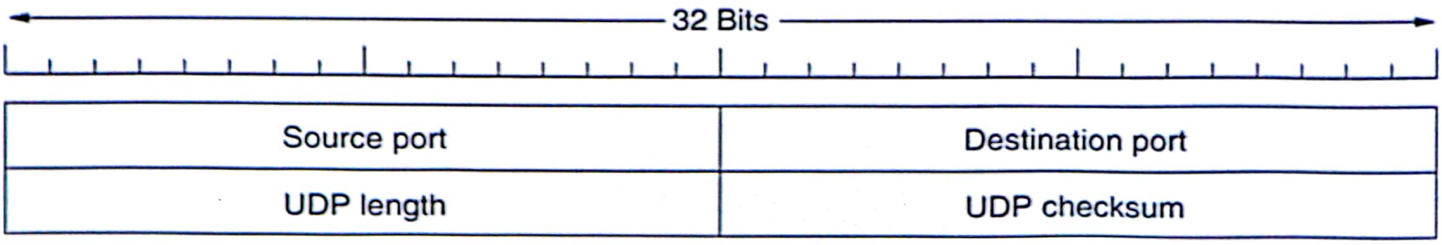
- **Primitives**

Primitive	Meaning
SOCKET	Create a new communication end point
BIND	Attach a local address to a socket
LISTEN	Announce willingness to accept connections; give queue size
ACCEPT	Block the caller until a connection attempt arrives
CONNECT	Actively attempt to establish a connection
SEND	Send some data over the connection
RECEIVE	Receive some data from the connection
CLOSE	Release the connection

Fig. 6-6. The socket primitives for TCP.

知识点：第6章

- Three-way handshake
- Connection Release



知识点：第6章

- Some well known ports：

Port	Protocol	Use
20, 21	FTP	File transfer
22	SSH	Remote login, replacement for Telnet
25	SMTP	Email
80	HTTP	World Wide Web
110	POP-3	Remote email access
143	IMAP	Remote email access
443	HTTPS	Secure Web (HTTP over SSL/TLS)
543	RTSP	Media player control
631	IPP	Printer sharing

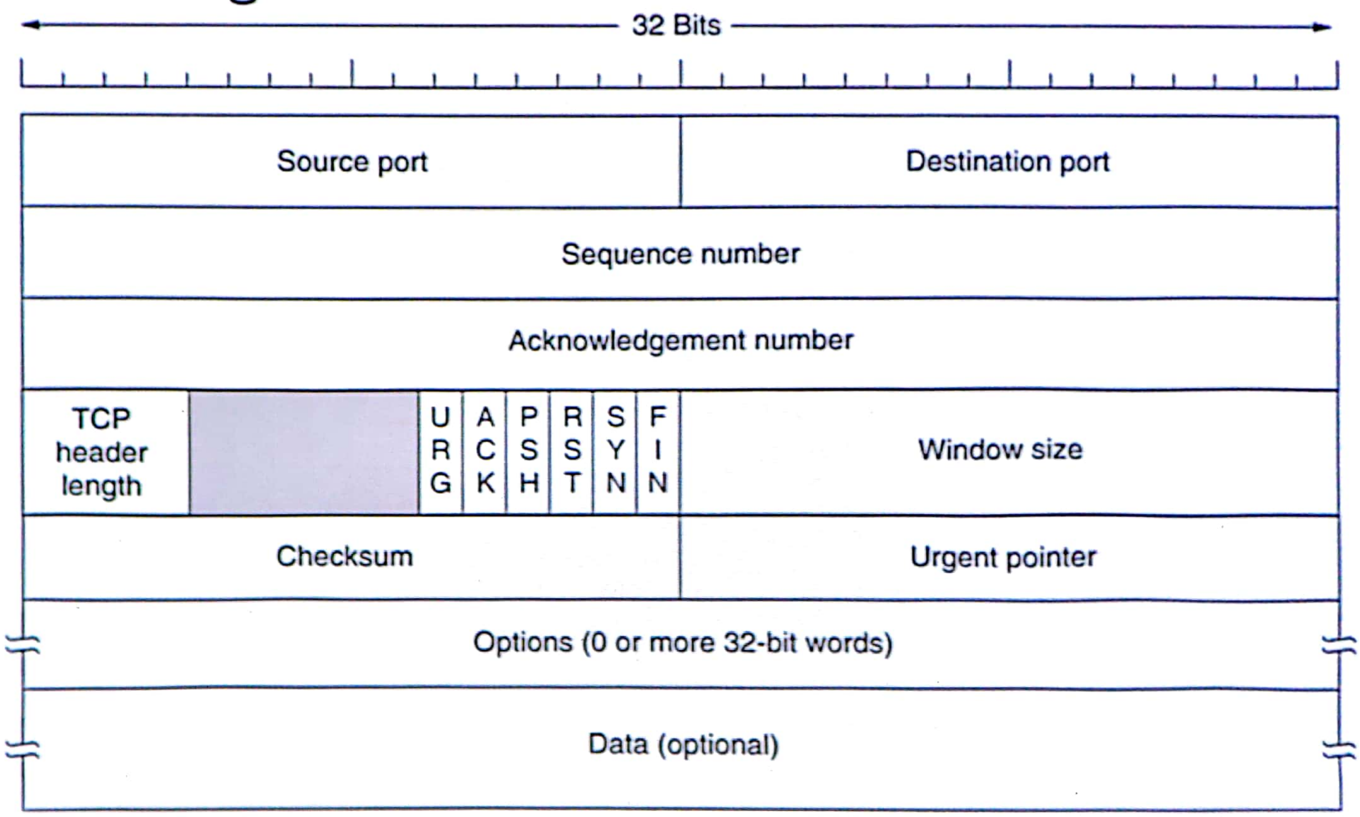
23 Telnet Remote login

知识点：第6章

- A TCP connection is a byte stream, not a message stream. Message boundaries are not preserved end to end.

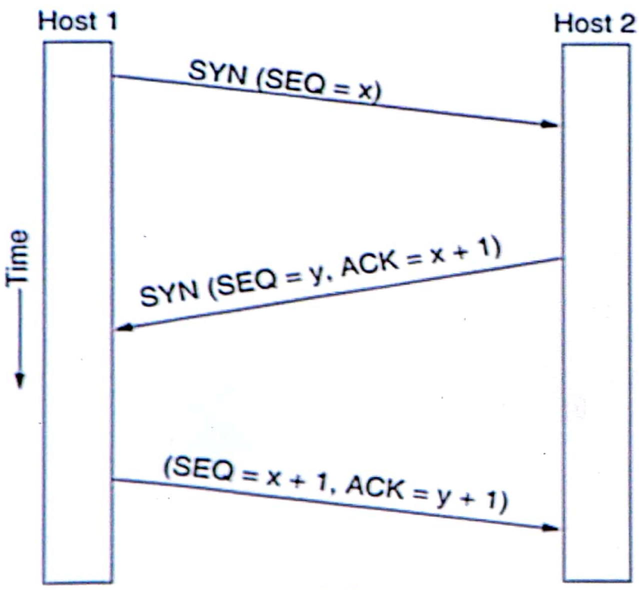
知识点：第6章

- TCP Segment



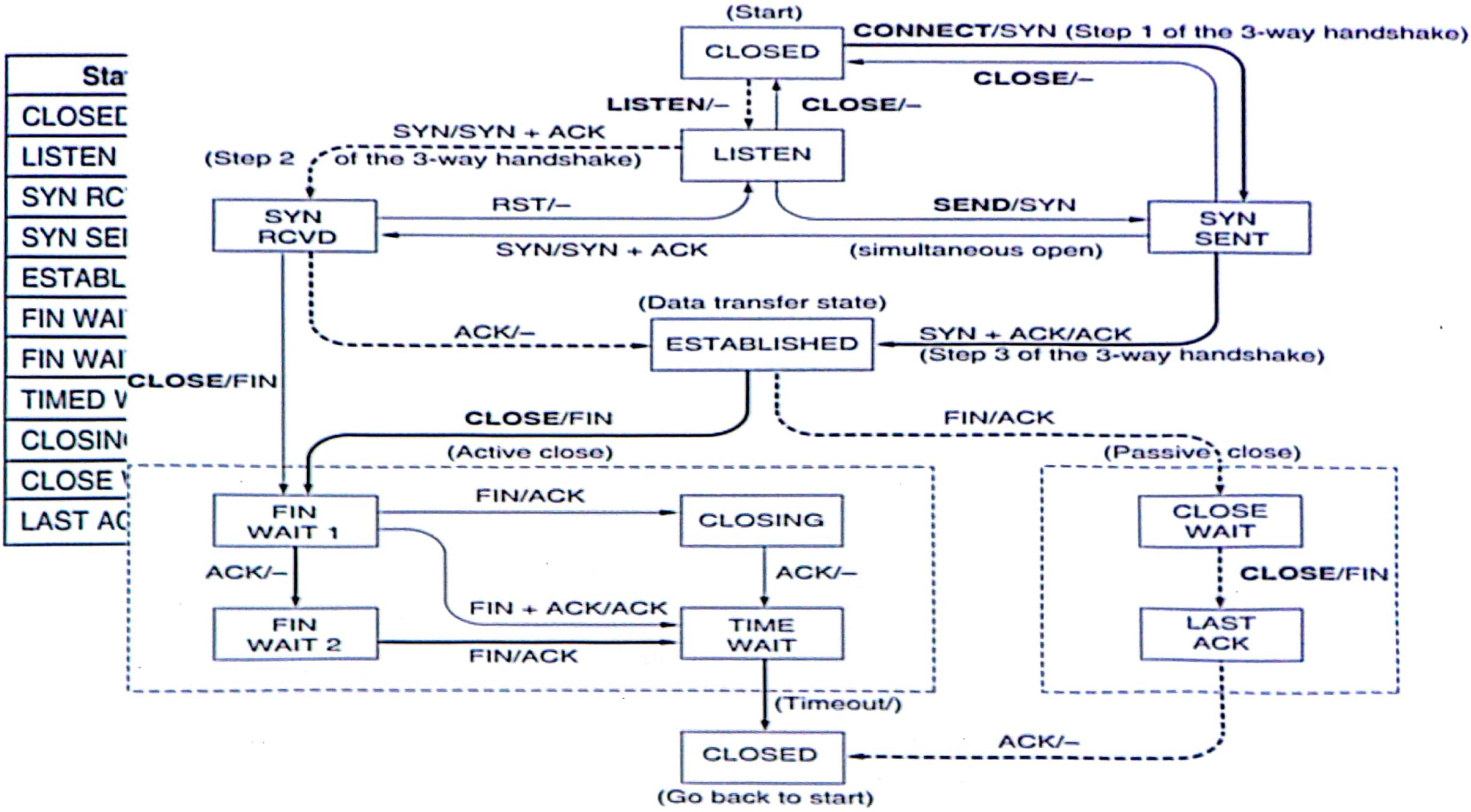
知识点：第6章

- TCP connection establishment in the normal case



(a)

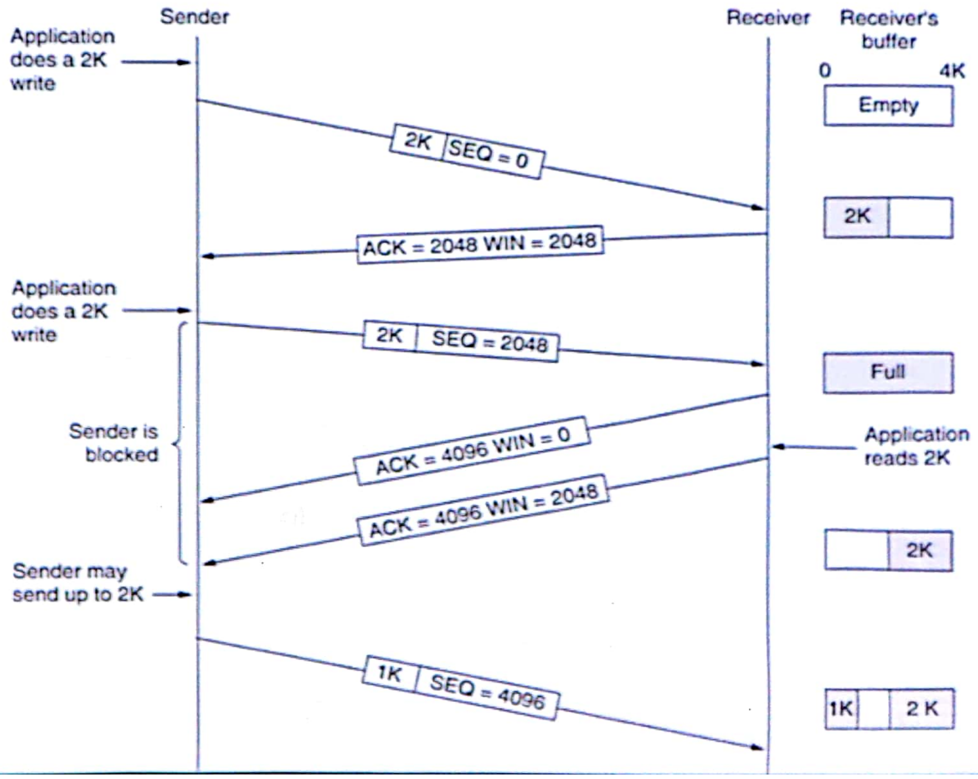
知识点：第6章



- 记住每个状态的含义

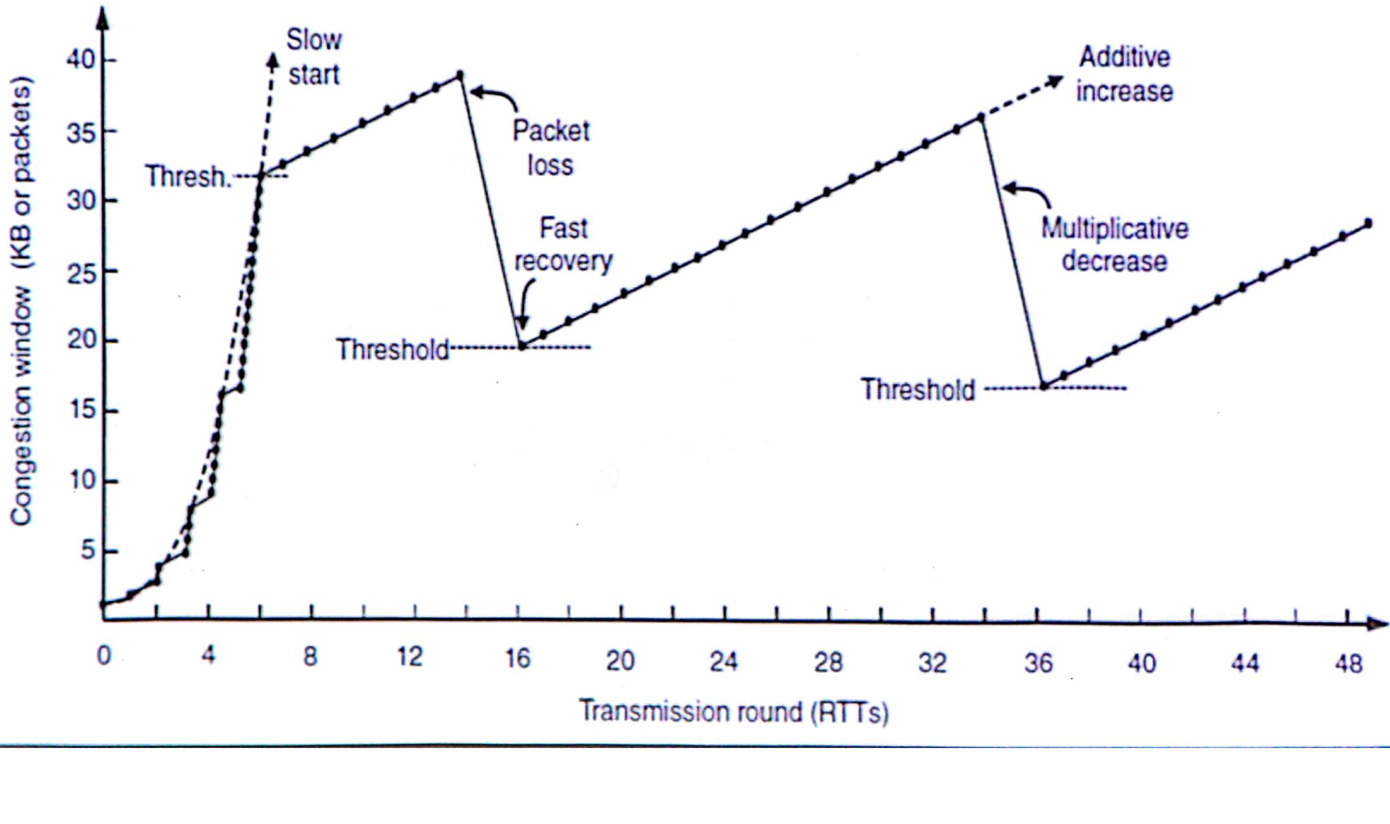
知识点：第6章

- Window management in TCP



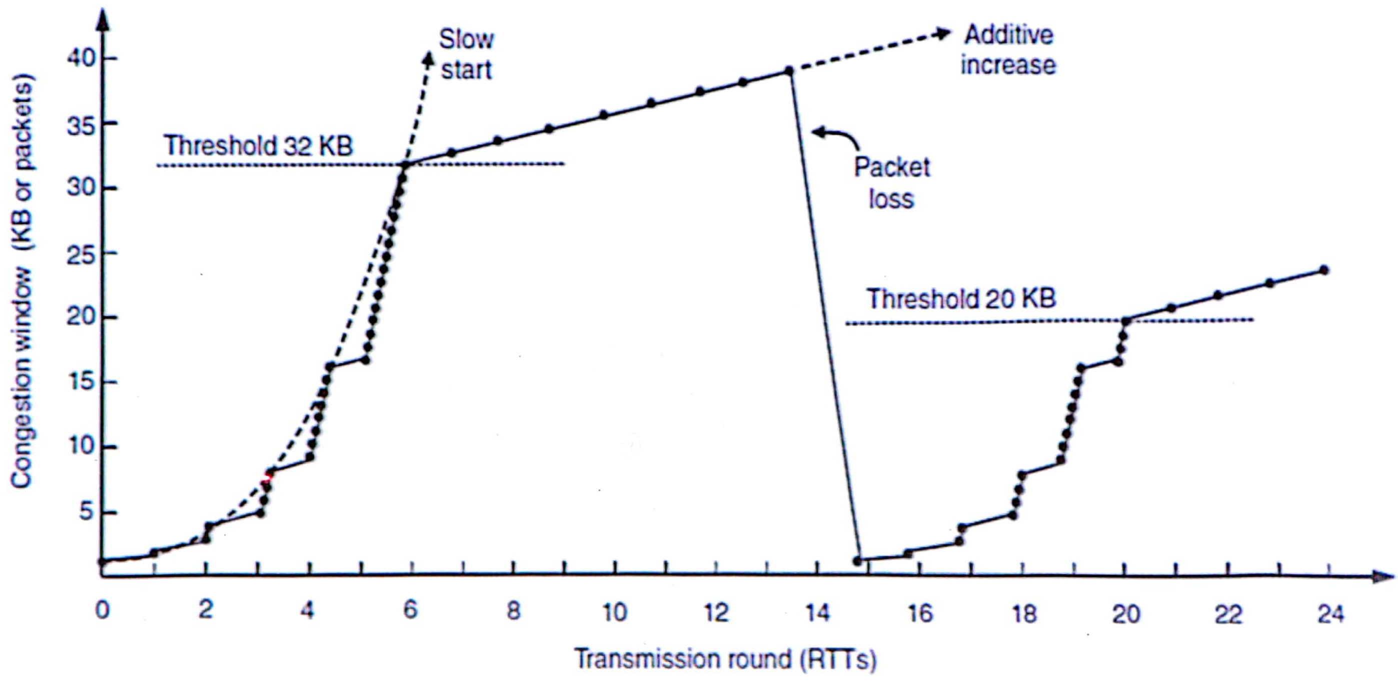
知识点：第6章

- Fast recovery and the sawtooth pattern of TCP Reno.



知识点：第6章

- Slow start followed by additive increase in TCP Tahoe



知识点：第6章

1. Which is the function of ports in the UDP datagram? The destination port is used to identify the application where to deliver payload

2. If the state of a TCP connection is in CLOSE_WAIT, what is the TCP entity waiting for? the application invoke a close primitive

3. If the window size field of the acknowledgment TCP segment is 30 KB, and the congestion window size is 40 KB, how many bytes could the sender transmit next time? 30 KB

4. Suppose that the TCP congestion window is set to 18 KB and a timeout occurs. How big will the window be if the next **four** transmission bursts are all successful? Assume that the maximum segment size is 1 KB. 8 KB

Host A continuously sends host B two TCP segments, which sequence number is 100 and 220. Please answer following 4 **questions**:

1、 How many bytes of data does the first segment bring?

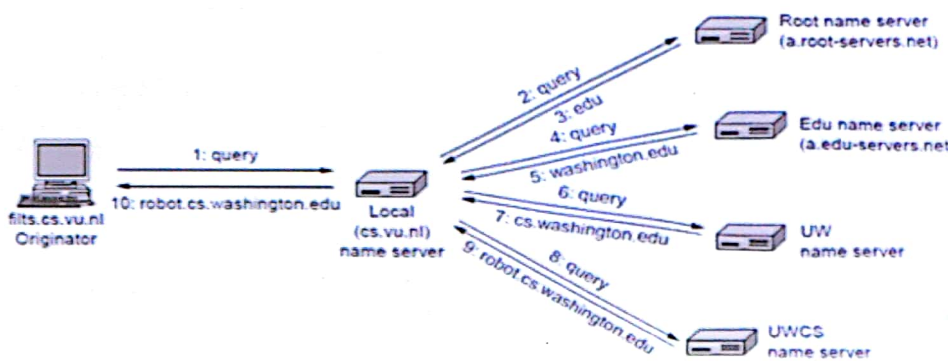
A. 99　　B. 100　　C. 120　　D. 220

知识点：第7章

- Network Application Model
- DNS
- Electronic Mail
- World Wide Web
- FTP [*]
- Multimedia [*]
- Content Delivery [*]

知识点：第7章

- C/S, P2P
- The DNS Name Space
  - DNS名字空间被划分为一些不交叉的区域(zone)
- Resource Records
  - The principal DNS resource records types
- Name Servers
  - Recursive Query
  - Iterative Query



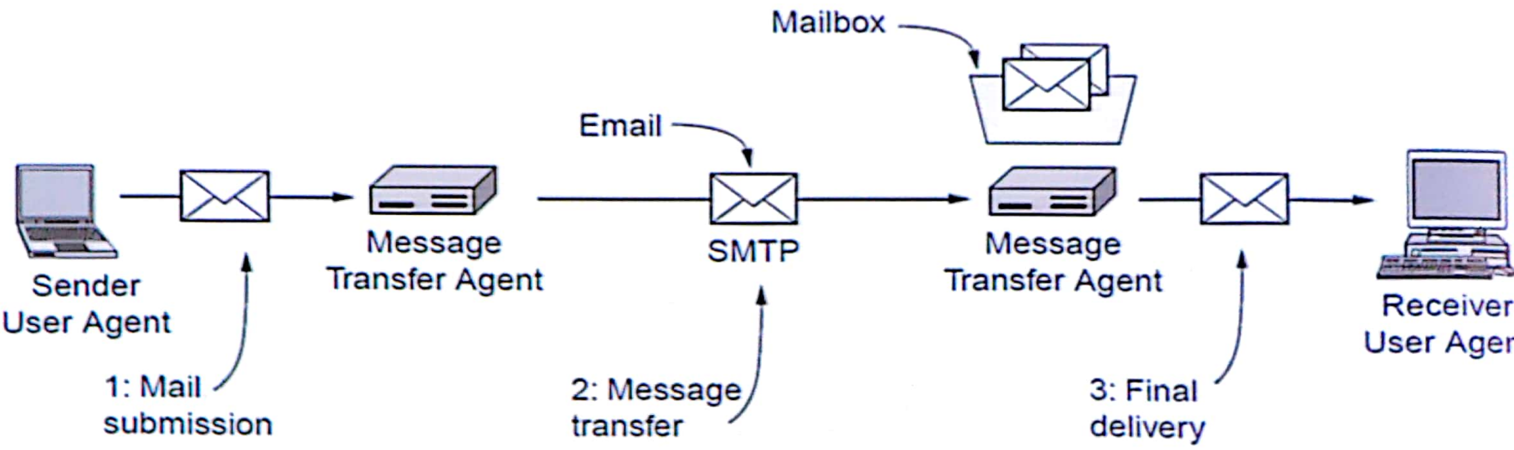
知识点：第7章

mail.xyz.com.cn	86400	IN	A	202.10.108.44
unix1.xyz.com.cn	86400	IN	A	202.10.108.45
smtp.xyz.com.cn	86400	IN	A	202.10.108.46
hr.xyz.com.cn	86400	IN	A	202.10.108.47
xyz.com.cn	86400	IN	MX	1 mail.xyz.com.cn
xyz.com.cn	86400	IN	MX	2 smtp.xyz.com.cn
hr.xyz.com.cn	86400	IN	MX	sun.xyz.com.cn
sun.xyz.com.cn	86400	IN	CNAME	unix1.xyz.com.cn

Q: What is the mail server address for zhang3@hr.xyz.com.cn?

知识点：第7章

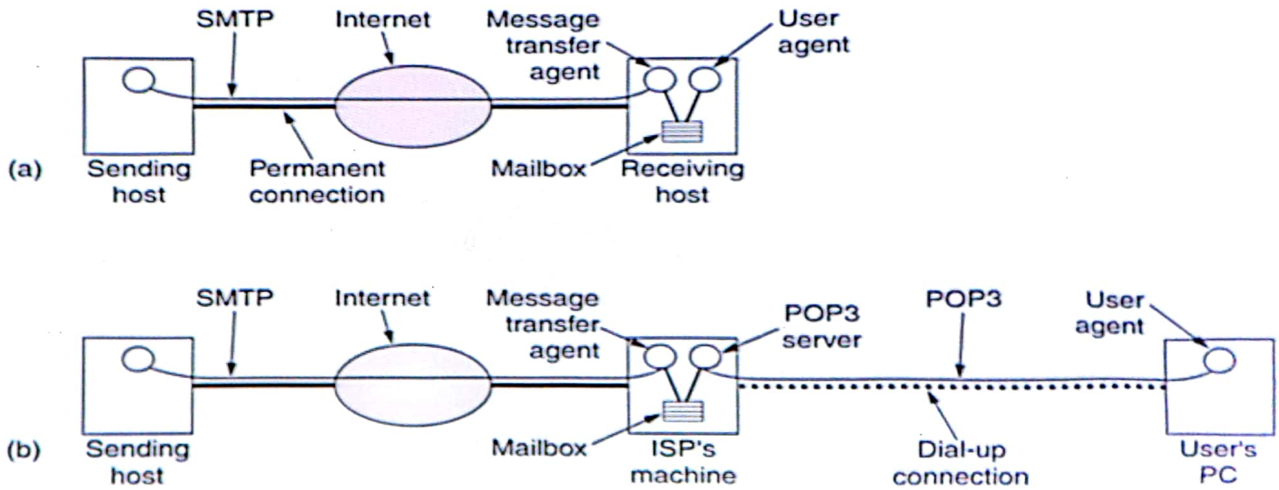
- EMAIL:



- 基本ASCII格式 和 多媒体扩展格式
- 消息：基本信封+头字段+空行+主体
- MIME – 多用因特网邮件扩展
  – Base 64：64个基本字符编码(24bit->4个6bit字符)

知识点：第7章

- SMTP – 简单邮件传输协议
  - 一个简单的ASCII协议
  - 端口号：25
  - TCP连接
- Final Delivery: POP3(110) vs IMAP(143)



知识点：第7章

- Page\Hypertext\browser\hyperlink\bookmark\hypermedia\TCP 连接 (port number: 80)
- URLconsists of 3 parts: the protocol (http), the DNS name of the host (www.zju.edu.cn), and the file name (index.html)
- Cookies
- Static/Dynamic Web Documents /CSS/XML/XSL
- HTML, common HTML tags
- XHTML
- Server-Side Dynamic(CGI\PHP\ASP\JSP)
- Client-Side Dynamic(Javascript\Java\ActiveX controls)
- Ajax

知识点：第7章

- HTTP
  - **Connections**
  - **Methods**
  - **Message Headers (parameters)**
    - Request headers
    - Response headers
  - **3 methods to improve network performances**
    - **Caching**
    - **Server replication (or called Mirroring)**
    - **Content delivery networks**

知识点：第7章

The built-in HTTP request methods：含义和操作细节

Method	Description
GET	Request to read a Web page
HEAD	Request to read a Web page's header
PUT	Request to store a Web page
POST	Append to a named resource (e.g., a Web page)
DELETE	Remove the Web page
TRACE	Echo the incoming request
CONNECT	Reserved for future use
OPTIONS	Query certain options

知识点：第7章

- **The status code response groups**

Code	Meaning	Examples
1xx	Information	100 = server agrees to handle client's request
2xx	Success	200 = request succeeded; 204 = no content present
3xx	Redirection	301 = page moved; 304 = cached page still valid
4xx	Client error	403 = forbidden page; 404 = page not found
5xx	Server error	500 = internal server error; 503 = try again later

知识点：实验

- 协议分析：IP header、TCP header
  - Wireshark中filter的使用格式
- 熟悉Cisco命令
- 二层配置、VLAN
- 路由配置：静态、OSPF、BGP